COMPUTER, HUMIDITY-TEMPERATURE CP-223( )/UM

1. SCOPE

1.1 This specification covers one type of circular slide rule for ascertaining temperature and humidity from data received from radiosonde transmissions, designated as Computer, Humidity-Temperature CP-223( )/UM. (See 6.2)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-591	Boxes, Fibreboard, Wood-Cleated.
PPP-B-601	Boxes, Wood, Cleated Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-117	Bags, Interior Packaging.
MIL-L-10547	Liners, Case, Waterproof.
MIL-B-13239	Barrier Material, Waterproofed, Flexible, all Temperature.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

SIGNAL CORPS

SC-DL-21182	Computer, Humidity-Temperature CP-223( )/UM.

FSC 6660

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications. - The following documents of the issue in effect on the date of invitation for bids form a part of this specification.

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the National Classification Board, 1424 16th Street, N. W., Washington 6, D. C.).

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification
Uniform Freight Classification

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.).

3. REQUIREMENTS

3.1 Parts, materials, and processes - general. - Parts, materials, and processes shall be such as to enable the equipment to meet its performance and other requirements as specified herein.

3.2 Construction. - The computer shall be constructed in accordance with the drawings listed on SC-DL-21182, with associated master plates (see 6.4), and the requirements specified herein.

3.3 Preproduction samples. - The contractor shall furnish preproduction samples for approval, if and as required by the invitation for bids and contract. (See 6.1(1)).

3.4 Technical literature. - Technical literature shall be furnished as specified in the bid request and contract.

3.5 Service conditions. - The computer shall meet the requirements of this specification under any probable combination of the following:

a. Temperature: Any ambient temperature in the range of -65°F to /130°F.

b. Elevation. Any elevation up to 50,000 feet above sea level.

c. Relative humidity. Any relative humidity up to 90 percent.

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3.6 Accuracy. - The computer shall meet the following check point requirements:

3.6.1 Temperature scale (base plate). - The temperature listed in column (2) below shall aline with the corresponding frequencies listed in column (1) below within an accuracy of ± 0.1 degree centigrade. (See 4.6.1)

(1) Recorder Marking Frequency Plate	(2) Temperature (C^O) Temperature Plate
71.2	+30.0
81.8	+60.0
53.2	0.0
29.7	-30.0
10.8	-60.0

3.6.2 Frequency (recorder) scale (center plate). - Relative humidity readings shall be obtained within the specified accuracy of 4.6.2 by aligning the temperature markings on the cursor with the recorder (frequency plate) markings when the frequency and humidity plates have been locked in place. (See 4.6.2)

3.7 Markings. - The lines, arcs, figures, divisions, and lettering on the temperature, frequency (recorder) and humidity plates, and the cursor shall conform to the applicable drawings listed on SC-DL-21182.

3.7.1 Lines and figures shall be sharply defined and clearly legible.

3.8 Surface finish. - All exposed surfaces of the computer shall be finished as specified in applicable drawings in Drawing and Data List SC-DL-21182.

3.9 Flatness. - Each plate (temperature, frequency and humidity) when laid on a flat surface, shall show no warpage greater than 1/16 inch at any point and the average shall be no greater than 1/32 inch. This requirement shall apply both before and after the plate is subjected to the tests of 4.5.

3.10 Interchangeability. - Corresponding units shall be physically and functionally interchangeable without modification thereof or of other items with which the units are used.

3.11 Workmanship. - The plastic plates, cursor, and parts shall be manufactured in a thoroughly workmanlike manner, in accordance with the applicable portions of the following paragraphs herein.

3.1 Parts, materials and processes - general
3.2 Construction
3.6 Accuracy
3.7 Markings
3.8 Surface finish
3.9 Flatness
3.10 Interchangeability

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor responsibility. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3.)

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.4 and 4.9:)

(1) Procurement inspection of items before preparation for delivery. (See 4.4.)

(2) Procurement inspection of preparation for delivery. (See 4.9 and section 5.)

4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I and the group A inspection specified in table II. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements.

Table I. - Preproduction inspection

Inspection (For additional preproduction inspection see 4.3)	Rqt Par.	Insp Par.
High temperature	3.5a	4.5.1
Low temperature	3.5a	4.5.2
Elevation	3.5b	4.5.3
Humidity	3.5c	4.5.1

4.4 Procurement inspection of items before preparation for delivery. - The contractor shall perform the inspection specified by (a) and (b) below, to demonstrate compliance with specified requirements. However, this does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the inspection specified in (a), to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection. Also see 6.3.)

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(a) Procurement inspection shall consist of group A inspection as specified in 4.4.1 through 4.4.2.

(b) When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.4.1 Group A inspection. - This inspection (including sampling) shall conform to table II and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table II - Group A Inspection

Inspection	Rqt. Par.	Insp. Par.	AQL Major	AQL Minor
Visual and Mechanical	3.11	4.7)		
Construction	3.2	4.7)		
Marking	3.7, 3.7.1	4.7)	1.0%	4.0%
Flatness	3.9	4.7)		
Interchangeability	3.10	4.8)		
Accuracy				
Temperature Plate	3.6.1	4.6.1)	1.0%	*
Frequency Plate	3.6.2	4.6.2)		

* All accuracy defects are of major category

4.4.2 Disposition of nonconforming product. - When defective sample units or failed inspection lots are submitted for acceptance, such products shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.5 Service condition tests. -

4.5.1 High temperature and humidity test. - The assembled computer shall be exposed, in an air circulating oven, to a temperature of $130^{\circ} \pm 5^{\circ}$F at 90 percent humidity for 24 hours. After the exposure, the computer shall be disassembled and the plates examined for compliance with 3.9, and there shall be no signs of discoloration or separation of the laminations.

4.5.2 Low temperature test. - The assembled computer shall be placed in a cold chamber and subjected to $-65^{\circ} \pm 5^{\circ}$F for a period of 24 hours. Following this exposure, the computer, shall be removed from the chamber and allowed to remain at normal room temperature for a period of 24 hours, after which it shall meet the requirements of 3.9, and there shall be no signs of discoloration, cracking, or separation of the laminations.

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4.5.3 Elevation. - The computer shall be placed in a pressure chamber and the pressure reduced to an absolute pressure of 3.4 inches of mercury, corresponding to an altitude of 50,000 feet. The pressure shall be maintained at this value for 4 hours, after which the computor shall be removed from the chamber and examined for compliance with 3.9 and shall show no signs of discoloration, cracking or separation of the laminations.

4.6 Accuracy. - The computer shall meet the following check point requirements:

4.6.1 Temperature plate. - Loosen the top and bottom nuts on the spindle of the computer; rotate the frequency plate (Recorder Markings) so that 71.2 is opposite 30° on the temperature scale (base plate) and lock in position by tightening the top nut. Check for compliance with 3.6.1.

4.6.2 Frequency-Humidity plates. - Loosen the top and bottom nuts on the spindle by hand; rotate the cursor to have the hairline line-up with recorder marking 80 on the frequency plate; rotate the humidity plate to have the 35 percent line (curve) intersect the hairline of the cursor at the $25^{\circ}C$ marking and lock in position by tightening the large nut; rotate the cursor to have the temperature marking listed in column 4 of the following table line-up with the relative humidity line listed in column 3 of the same table. The corresponding recorder marking as listed in column 1 for each setting shall be obtained within the specified accuracy (col. (2)).

(1) Recorder Marking (Frequency Plate)	(2) Specified Accuracy	(3) Relative Humidity Line Humidity Plate	(4) Temperature Marking
11.8	±0.2	95	±25
34.7	±0.2	80	±25
60.7	±0.2	65	+25
46.8	±0.2	75	+35
13.6	±0.2	90	0
82.9	±0.2	20	0
53.0	±0.3	60	-40
79.5	±0.3	35	-40

4.7 Visual and mechanical inspection of equipment. - The plates, cursor, and parts such as the spindle, washers and nuts shall be inspected for workmanship, mechanical fit, and miscellaneous defects. Fastening nuts shall be inspected for mechanical operation.

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Table III - Classification of Visual and Mechanical Defects

Classification	Defect
Major	Loose mechanical operation Illegible or distorted markings Excessive warpage Air bubbles in plastic Loose laminations Cracks in plastic Scratches in plastic Mounting holes off center Excessive discoloration Pencil marks on finished surfaces difficult to erase
Minor	Minor discoloration Incomplete markings Mechanical Operation too stiff Chips in edges of plates

4.8 Inspection for interchangeability. - The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.10. When a listed dimension is not within the design limits set forth in the applicable drawing (see SC-DL-21182) it shall be considered a major defect.

(a) External and internal dimensions of plates and cursor.

(b) Thread dimensions of lock nuts.

(c) Dimensions of spacing washers.

4.9 Procurement inspection of preparation for delivery. - The contractor shall perform the inspection specified by 4.9.1 and 4.9.2 to demonstrate compliance with specified requirements. However, this does not relieve the contractor of responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the inspection specified in 4.9.1 and 4.9.2 to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection. Also see 6.3.) When Standard MIL-STD-105 specified action by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

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4.9.1 Preservation and packaging. - Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown on table IV.

4.9.2 Packing and marking of exterior containers. - Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on table V.

4.9.2.1 Inspection lot. - A lot of visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.9.2.2 Disposition of nonconforming product. - Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.10 Rough handling test (preparation for delivery). - When rough handling test is required by the contract (see 6.1(d)(2)e., the following test shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Visual and Mechanical.

Table IV - Classification of defects for preservation, packaging and marking

Major	Minor
1. Use of improper or defective material.	1. Item not proper blocked or braced within the unit package to prevent movement.
2. Quantity in unit package not as specified.	
3. Incorrect packaging method applied.	
4. Cushioning (padding)omitted.	2. Packaging material damaged.
5. Cushioning (padding) inadequate for the protection of the barrier material from projections, sharp edges, or other similar features of the item.	3. Any item of marking information other than 8 thru 10 omitted, incorrect, or illegible.
6. Cushioning inadequate for the physical and mechanical protection of the item.	
7. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.	

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8.	Stock number omitted; incorrect, or illegible.	
9.	Nomenclature omitted, incorrect, or illegible.	
10.	Marking of quantity of items in package omitted, incorrect or illegible.	

Table V - Classification of defects for packing and marking

	Major		Minor
1.	Use of improper or defective material.	1.	Defective taping or sealing of carton.
2.	Quantity in pack not as specified.	2.	Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
3.	Gross weight in excess of specified amount.	3.	Any item of required marking information other than 12 thru 19 listed under major defects omitted, incorrect, or illegible.
4.	Case liner omitted.		
5.	Case liner improperly sealed, punctured, or damaged.		
6.	Box closure not as specified.		
7.	Type, grade, class, and style of the shipping container not as specified.		
8.	Strapping omitted.		
9.	Strapping inadequate or incorrectly applied.		
10.	Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.		
11.	Shipping documents or packing list omitted.		
12.	Stock number omitted, incorrect, or illegible.		
13.	Nomenclature omitted, incorrect, or illegible.		
14.	Marking of quantity of items in pack, omitted, incorrect, or illegible.		
15.	Destination marking omitted, incorrect, or illegible.		
16.	Service designation (color marking) omitted.		
17.	Special marking or labeling (when required) omitted.		
18.	Overseas code marking omitted, incorrect, or illegible.		
19.	Unsealed carton.		

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5. PREPARATION OF DELIVERY

5.1 Preservation and packaging. -

5.1.1 Level A. - Computer, Humidity-Temperature CP-223( )/UM shall be preserved packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and as described herein.

5.1.1.1 Computer, Humidity-Temperature and technical literature. - Each computer and each technical literature shall be packaged Method IC-8, as follows: Cushion each computer and each technical literature together between two sheets of type III, single face, flexible corrugated fiberboard. Inclose the cushioned computer and technical literature within a close-fitting, type I, class b bag, conforming to Specification MIL-B-117. Heat seal to effect closure.

5.1.1.2 Intermediate container. - Place a quantity of unit packages as specified in 5.1.1.1 together, within a close-fitting, folding, paperboard box, style II, type A, class a, conforming to Specification PP-B-566. Closure shall be in accordance with the requirements of the appendix of the box specification. The gross weight of any intermediate container shall not exceed five pounds.

5.1.2 Level C. - Computer, Humidity-Temperature CP-223( )/UM shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage, during direct shipment to the first receiving activity.

5.2 Packing. -

5.2.1 Level A. - A quantity of computers, packaged as specified in 5.1.1.2 shall be packed for shipment within close fitting wirebound wood; wood cleated fiberboard; wood cleated plywood, or nailed, wood, box, conforming to the requirements of Specification PPP-B-585, class 3, PPP-B-591 (overseas type), PPP-B-601 (overseas type) or PP-B-621, class 2, respectively. Fabricate the box to fit the contents snugly. Shipping containers shall be fitted with a waterproof case liner of barrier material, grade B, type CW-2 conforming to Specification MIL-B-13239 and fabricated in accordance with Specification MIL-L-10547. Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of any shipping container shall not exceed 200 pounds.

5.2.1.1 Metal strapping. - Shipping containers shall be strapped in conformance with the requirements of the appendix of the container specification.

5.2.2 Level B. - Computer, Humidity-Temperature CP-223( )/UM shall be packed as specified in 5.2.1 except that the boxes shall be domestic type or class.

5.2.3 Level C. - Computer, Humidity-Temperature CP-223( )/UM shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification No. 4 and Consolidated Freight Classification, Rule 21 for rail shipment, National Motor Freight Classification No. A-3 and No. 14 for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed at lowest transportation rate.

5.3 Marking. - Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

(c) The specific paragraphs of section 5 which are applicable to the particular procurement.

(d) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental test can be completed on one sample while complete performance measurements can be made on the second sample. (See 3.3).

(2) Preproduction pack(s) as follows:

a. Makeup of pack(s).
b. Number of each kind of pack to be submitted.
c. Inspection to be performed thereon--including rough handling test, which will not be performed as procurement inspection.

d. Five copies of process sheet, using format shown on Drawing SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.

e. A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required. (See 3.4)

6.2 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: CP-223W/UM. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

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6.3 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

6.4 Queries concerning availability of master plates should be addressed to Commanding Officer, U. S. Army Signal Equipment Support Agency, Ft. Monmouth, New Jersey.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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